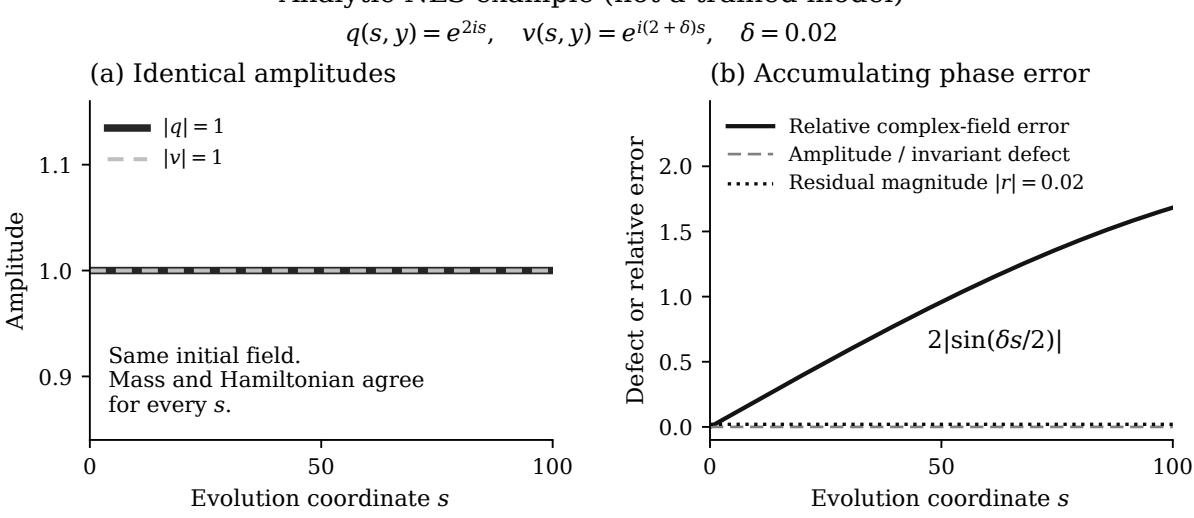
Analytic NLS example (not a trained model)
$q(s,y) = e^{2is}, \quad \nu(s,y) = e^{i(2+\delta)s}, \quad \delta = 0.02$
(a) Identical amplitudes
$|q| = 1$
$|\nu| = 1$
Amplitude
1.1
1.0
0.9
Same initial field.
Mass and Hamiltonian agree
for every s.
0
50
100
Evolution coordinate s
(b) Accumulating phase error
Relative complex-field error
Amplitude / invariant defect
Residual magnitude $|r| = 0.02$
Defect or relative error
2.0
1.5
1.0
0.5
0.0
$2|\sin(\delta s/2)|$
0
50
100
Evolution coordinate s
Exact formulas for $iq_s + q_{yy} + 2|q|^2 q = 0$ on a finite periodic interval.